



J. Dairy Sci. TBC

<https://doi.org/10.3168/jds.2024-24923>

© TBC, The Authors. Published by Elsevier Inc. on behalf of the American Dairy Science Association®.
This is an open access article under the CC BY license (<https://creativecommons.org/licenses/by/4.0/>).

Graduate student literature review: Integrating concepts of animal welfare and health-related quality of life for pre-weaned dairy calves

Jean Ramos,^{1*} Marianne Villettaz-Robichaud,¹ Jocelyn Dubuc,¹ Débora Santschi,² Jean-Philippe Roy,¹ Gilles Fecteau,¹ and Sébastien Buczinski^{1§}

¹Département des Sciences Cliniques, Faculté de Médecine Vétérinaire, Université de Montréal, 3200 rue Sicotte, St-Hyacinthe, QC, J2S 2M2, Canada.

²Lactanet, 555 Boul des Anciens-Combattants, Sainte-Anne-de-Bellevue, QC H9X 3R4, Canada.

ABSTRACT

Dairy calf welfare assessment tools focusing on the pre-weaning period have been proposed in recent research. Despite the existence of these tools, assessing the welfare and health-related quality of life (HRQoL) of dairy calves remains challenging. These difficulties may stem from the complexity of assessing all dimensions of calf welfare and the validity, reliability, and feasibility of the indicators used in assessment tools. This review aims to discuss welfare and HRQoL concepts and integrate them into a framework to facilitate the understanding of dairy calf welfare and HRQoL. The review also identifies on-farm dairy calf welfare indicators and explores how their validity, reliability, and feasibility have been evaluated. HRQoL, as a component of the quality of life concept, is used to determine how an animal feels during illness using behavioral expressions of affective states. In this review, we adapted a human HRQoL framework for dairy calves, illustrating the interconnection of 3 domains of animal welfare (behavior, mental state and health) in assessing calves' HRQoL. It is worth noting that a limited number of HRQoL assessment tools have been developed for dairy calves, and there is no standard way to assess the welfare of pre-weaned dairy calves. Some studies focus on specific aspects of animal welfare, while others address this concept more broadly. While behavioral indicators have been explored in dairy science, they often remain disconnected from the concept of HRQoL. After reviewing various welfare assessment tools focusing on pre-weaned dairy calves, 44 welfare indicators were selected. Considering the selected welfare indicators, we observed that their validity, reliability, and feasibility have not been extensively explored. This review contributes to understanding welfare and HRQoL

concepts for pre-weaned dairy calves and highlights opportunities to enhance the assessment of welfare and HRQoL for these animals.

Key words: Well-being, On-farm assessment tools, Welfare indicators

INTRODUCTION

Animal welfare continues to be an ethical concern for pre-weaned dairy calves. Ensuring good animal welfare involves considering the well-known 5 freedoms (Farm Animal Welfare Council, 1993). Calves need adequate nutrition, a suitable environment, good health, freedom to express behaviors, and a contented mental state (Mellor and Reid, 1994). To protect and improve dairy calves' welfare, it is important to assess it on a farm and identify areas for improvement. On-farm assessment tools are usually used for this purpose, and various approaches and indicators have been previously identified (Vasseur et al., 2010; Barry et al., 2019; Hayer et al., 2021).

The Welfare Quality® project organized welfare indicators according to 12 criteria and 4 principles (i.e., good feeding, good housing, good health, and appropriate behavior; Blokhuis et al., 2010). Other assessment tools differentiate welfare indicators according to animal-based (indicators that can be evaluated by direct or indirect observation of the animals) and resource-based (e.g., presence of concentrate and water; Barry et al., 2019; Hayer et al., 2021). Calves' welfare indicators were also described according to management areas, such as calving, colostrum, painful procedure, calf feeding, weaning, and housing (Vasseur et al., 2010).

Despite the existence of these tools, the different approaches and indicators can pose a challenge in choosing which tool to use and which welfare indicators an assessor should prioritize. The variability in approaches may result from varying levels of concern for animal welfare (Fraser et al., 1997). For example, behavioral indicators were not included in the welfare assessment tool for

Received March 15, 2024.

Accepted December 21, 2024.

*Corresponding author: jean.silva.ramos@umontreal.ca

§ Advisor: s.buczinski@umontreal.ca

The list of standard abbreviations for JDS is available at adsa.org/jds-abbreviations-24. Nonstandard abbreviations are available in the Notes.

dairy calves proposed by Hayer et al. (2021). However, indicators such as social play and abnormal oral behaviors were selected in the Welfare Quality® assessment protocols for veal calves and another assessment tool for dairy calves (Barry et al., 2019). A recently developed tool for dairy calves focuses on Health-Related Quality of Life (HRQoL) by evaluating the animals' experiences through behavioral expressions during respiratory disease episodes (Bell et al., 2023, 2024). Thus, behavioral indicators for dairy calves have been used in different ways to assess animal welfare and HRQoL. Although the concept of HRQoL has been discussed (Bell et al., 2023, 2024), it is unclear how to differentiate the use of behavioral indicators in assessing animal welfare and HRQoL. This lack of clarity provides an opportunity to discuss and integrate these concepts. An interesting framework was developed in human science to better understand HRQoL, its domains, and its outcomes (Testa and Simonson, 1996). Adapting this framework for calf HRQoL and welfare could be beneficial to understanding these topics.

In addition to conceptual understanding, it is essential to assess parameters such as the validity, reliability and feasibility of dairy calf welfare indicators in an on-farm assessment (Knierim and Winckler 2009). Validity refers to the capacity of an indicator to assess what it is supposed to (de Graaf et al., 2017). A reliable indicator must have good inter- and intra-rater reliability (Harvey, 2021). Inter-rater reliability refers to the consistency with which different raters yield the same results when assessing the same indicator. Intra-rater reliability refers to the consistency of evaluations of an indicator by the same rater within a short period (Harvey, 2021). Feasibility is generally associated with the time and cost needed to evaluate an indicator (Knierim and Winckler, 2009). While these parameters have been explored, a recent review highlighted the need for more studies evaluating these parameters for animal welfare (Hampton et al., 2023). The principles of validity, reliability, and feasibility need to be considered when selecting optimal welfare indicators for routine use on-farm.

This review aims to achieve 2 primary objectives. The first objective is to discuss welfare and HRQoL concepts and integrate them into a framework based on a human HRQoL approach (Testa and Simonson, 1996). The second objective is to identify on-farm dairy calf welfare indicators and explore how their validity, reliability, and feasibility have been evaluated.

ANIMAL WELFARE AND QUALITY OF LIFE (QOL) CONCEPTS

Animal welfare is a complex and multi-faceted concept (Lund et al., 2006). To date, consensus on the defini-

tion of animal welfare has not been reached (Reimert et al., 2023). Different conceptions of animal welfare have been defined according to the different concerns of society (Fraser et al., 1997). Three major areas of concern were proposed: basic health and functioning, natural living, and affective states (Fraser, 2008). The basic health and functioning concern is related to the freedom from diseases and injuries. Natural living concern refers to the need for calves to express natural behaviors and have access to natural elements. The term “affective states” relates to the concern that the animals need to feel well and describes how the animals are feeling (Fraser, 2008).

The Five Domains Model for animal welfare assessment, initially formulated in 1994, was designed to evaluate potential welfare compromises caused by experiments on sentient animals (Mellor and Reid, 1994). This model has been explored and updated for over 25 years (Mellor et al., 2020). The 5 domains are nutrition, environment, health, behavior, and mental state (Mellor et al., 2020). Animal welfare is assessed by considering indicators from the first 4 domains, all of which relate to the animal's experiences in the mental domain. For example, restricted water intake may be associated with the negative affect of thirst. The presence of an injury may be linked to the negative affect of pain. Restrictions on interactive activities could relate to feelings of loneliness or depression.

Quality of life (QoL) is a concept recently explored in animal science (Webb et al., 2019; Bell et al., 2023). More specifically, Yeates (2016) defined animal QoL as “an intrinsically evaluative concept of how valuable (positive or negative) each animal's life is for that animal from the animal's point of view.” HRQoL, as a component of the QoL concept, is used to determine how animals feel (from the animals' perspectives) during illness. Although assessing an animal's perspective or experience can be complex, behavioral expressions offer a useful alternative for this type of assessment (Yeates, 2016). For example, in an HRQoL tool for calves, behavioral expressions such as vigor and motivation at feed were selected to assess the calf's perspective in cases of respiratory disease (Bell et al., 2023).

When using the Five Domains Model, behavior provides insight into an animal's mental experience. In HRQoL assessments, behavior provides insight into an animal's mental experience during illness. The HRQoL tools link 3 domains of the Five Domains Model: behavior, mental state, and health. These tools play an essential role in human medicine, primarily helping with decisions regarding different types of treatment in clinical studies and clinical practice (Guyatt et al., 2007). The potential of this approach has been observed and explored in recent studies with dairy calves (Bull et al., 2021; Bell et al., 2023, 2024).

HEALTH-RELATED QUALITY OF LIFE (HRQOL) IN HUMANS AND ANIMALS

In a human approach to defining HRQoL outcomes (Testa and Simonson, 1996), 3 health domains were identified (physical, psychological, and work performance). Some questions were proposed to evaluate each domain. For example, functionalities in the physical domain can be assessed according to the ability of a person to perform physical activities. Questions about symptoms such as weakness, heartburn, weight gain, and thirst experienced during the past month have been included in this domain. In the psychological domain, questions about depression, anxiety, and other emotional states were selected. Work performance was evaluated according to how a person feels about their job. As an important observation, in this human methodology, thirst was included in the physical health domain (Testa and Simonson, 1996), and emotions, happiness, depression and anxiety were included in the psychological domain. In animal welfare, thirst, hunger, anxiety, fear, pain, and distress are included in the same classification, mental domain, or affective states (Mellor et al., 2020). As in human HRQoL, organizing affective states according to health domains could facilitate understanding of animals' HRQoL.

Bringing this human HRQoL approach to calves, we could suggest that calf HRQoL comprises indicators from health's physical and psychological domains. However, selecting indicators to assess how calves feel in relation to work performance, i.e., domain related to the usefulness of their life, is a more complex task. The usefulness of human life represents “the view that human quality of life has to do with higher values” (Webb et al., 2019). Considering that calves are intended for either milk or beef production, evaluating their feelings based on their human-proposed usefulness involves various stages, from early life to their productive lifespan (Reimert et al., 2023). This assessment may be challenging for calves, given the stages of life mentioned. Consequently, the domain of human work performance related to the concept of usefulness has been excluded from this review for calves. Further research is needed to explore and clarify all aspects related to the usefulness of calves and how they feel regarding this domain of life.

Using the human HRQoL framework as a reference (Testa and Simonson, 1996), we might propose that the assessment of calf HRQoL involves evaluating how a calf feels through its behavioral expressions of affective states, i.e., assessing the calf's perspective (Figure 1) according to its health status. The calf's perspective refers to a dimension known as ‘subjective perception’ (x-axis in Figure 1). For example, depending on health status, one calf could be alert and responsive, while another might appear dull, depressed, and less responsive (Barry

et al., 2019). Calves may experience varying subjective perceptions (ranging from low to high) of their health status. The health status is represented by the dimension ‘objective health status’ (y-axis in Figure 1). For example, a calf with respiratory disease and fever with treatment or a calf with respiratory disease without fever and no treatment. A low objective health status refers to a poor health condition, while a high objective status refers to a state of good health. Calves with similar health status may have different subjective perceptions of their health status. This variation in responses emphasizes the importance of assessing HRQoL individually. An assessment of the calf's perspective translates its health status into the actual HRQoL experienced (represented by the x and y coordinates in Figure 1). This assessment can consider different health domains, such as physical and psychological.

The HRQoL can be assessed using different indicators of behavioral expressions of affective states. In the framework (Figure 1), the Z refers to the various indicators that could be included in the evaluation (an ensemble of indicators to evaluate a given domain or HRQoL as a whole). For example, motivation at feed is a behavioral indicator related to the state of hunger and is used to assess how a calf feels in cases of respiratory disease (Bell et al., 2023). This indicator could be included in the physical domain of health due to its relationship with physical needs. The indicators ‘vigor’ and ‘responsiveness’ are behavioral indicators that can be used to assess whether a calf feels depressed in cases of respiratory disease (Bell et al., 2023). These indicators are related to the psychological status of a calf.

Unfortunately, few tools are currently available to assess the HRQoL of calves, making it challenging to include various indicators across different health domains. The recent study exploring HRQoL for dairy calves with respiratory disease considered all behavioral indicators within a single domain: behavioral expressions (Bell et al., 2023). Future research could focus on developing HRQoL tools for calves, integrating behavioral indicators of affective states related to other specific diseases and general health status.

ASSESSMENT OF CALF WELFARE AND HEALTH-RELATED QUALITY OF LIFE (HRQOL)

Some recent on-farm dairy calf welfare assessment tools have organized the welfare indicators into animal-based and resource-based categories (Barry et al., 2019; Hayer et al., 2021). For example, the animal welfare assessment tool proposed by Hayer et al. (2021) includes 14 animal-based indicators and 7 resource-based indicators. The Welfare Quality® assessment protocol for veal calves organizes the indicators according to 4 welfare

principles and 12 criteria (Blokhuys et al., 2010). It stated that animal welfare assessment could be evaluated by considering good feeding, good housing, good health, and appropriate behavior (Blokhuys et al., 2010). In another study, important areas for animal welfare - such as calving management, care for newborn calves, painful procedures, colostrum management, calf feeding, weaning, and calf housing - were identified, and an advisory tool was developed for dairy calves in Québec, Canada (Vasseur et al., 2010). In this tool, numerous resource- and management-based indicators were identified.

The importance of assessing calves' experiences during disease events was highlighted in a study aiming to develop an HRQoL tool for calves with respiratory disease (Bell et al., 2023). This tool incorporates indicators from 2 domains: clinical signs and behavioral expression (Bell et al., 2023). Seven indicators were included in the clinical signs' domain: respiratory effort, cough, ear carriage, eye appearance (discharge), eye appearance (vibrancy), nasal discharge, and body and head posture. The behavioral expression domain included 6 indicators:

vigor, motivation at feed, the volume of feed intake, movement to feed, spatial proximity, and responsiveness.

It is worth noting that only a limited number of HRQoL assessment tools have been developed for dairy calves, and there is no standard way to assess the welfare of pre-weaned dairy calves. Considering the different approaches and tools, it is important to understand how these approaches differ and how these tools could help in the assessment of calves' welfare. Some studies focus on the assessment of specific aspects of animal welfare (Vasseur et al., 2010; Hayer et al., 2021), while others address this concept more broadly (Blokhuys et al., 2010).

INTERCONNECTIONS BETWEEN CALF WELFARE INDICATORS

Another crucial aspect of understanding calf welfare involves recognizing the interconnections among welfare indicators. We used bovine respiratory disease (BRD) to exemplify how these indicators may be associated. Most studies are observational and reveal positive and negative associations between calf welfare indicators from different domains, but they do not indicate causality.

Indicators of the living environment may be linked to health status. For example, a calf nesting score (measuring calf bedding quantity and quality) has been associated with a reduced prevalence of respiratory disease (Lago et al., 2006), perhaps by lowering the potential negative effects of low temperatures and air drafts. Abrupt weaning and transportation are important stressors that can be related to the occurrence of respiratory disease (Griebel et al., 2014). Stressed calves may exhibit a deficient immune system, influencing the occurrence of respiratory disease (Hulbert and Moisés, 2016). Additionally, calves exposed to high temperatures may experience heat stress, contributing to the occurrence of respiratory disease (Louie et al., 2018).

The occurrence of BRD can also be associated with behavior-based measurements. For example, the presence of BRD has been associated with a behavioral score, which considers lying and standing postures, isolation, approach testing, and encouragement to rise (Cramer et al., 2016). Feeding behaviors have also been shown to be altered in sick calves (Conboy et al., 2021). In recent research, behavioral indicators have been used indirectly to detect calves with diarrhea and BRD (Cramer et al., 2019; Guevara-Mann et al., 2023).

Other associations between the domains may exist, and the ones reported were emphasized merely as common examples according to the literature. It is important to understand that indicators from different domains can be associated, and they can be used either directly (i.e., to assess the status of the domain for which the indicator was selected) or indirectly (i.e., to assess the status of

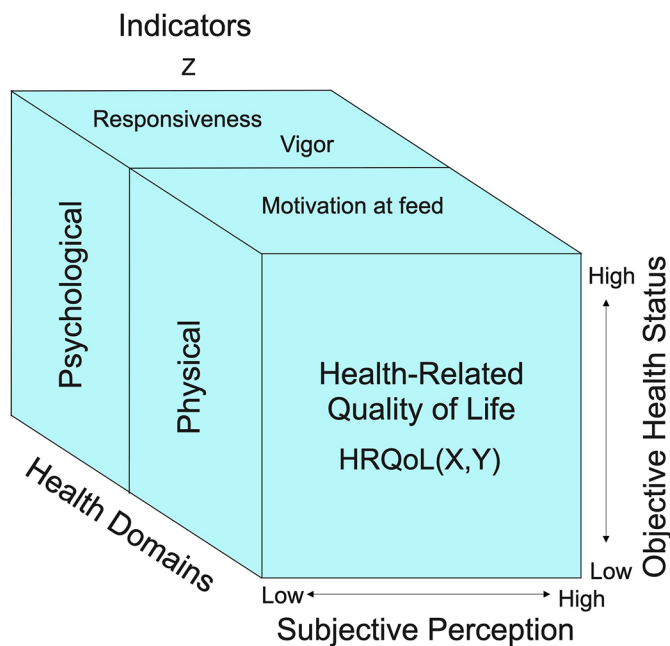


Figure 1. – The calf's subjective perception, assessed using behavioral indicators of affective states, translates its health status into the actual health-related quality of life (HRQoL) experienced - an adapted conceptual framework based on a human HRQoL approach (Testa and Simonson, 1996). The x-axis represents the calf's subjective perception (i.e., how it feels in relation to its health status). The y-axis represents the objective health status (i.e., the health condition of a calf). Coordinates x and y represent the actual HRQoL. The Z refers to the different indicators (e.g., behavioral expressions) that could be included in the HRQoL assessment (an ensemble of indicators to evaluate a given domain, such as physical and psychological, or HRQoL as a whole).

another domain). These interconnections highlight the importance of the link between the different welfare indicators and the specific link between behavior and health, which is important in assessing HRQoL. While behavioral indicators have been explored in dairy science, they often remain disconnected from the concept of HRQoL.

ON-FARM CALF WELFARE INDICATORS

Selection of on-farm calf welfare indicators

Different approaches have been used to select indicators in welfare assessment tools. A challenge in assessing welfare is knowing which indicators should be selected for a calf welfare assessment. Considering this obstacle, we identified and selected dairy calf welfare indicators in this section.

The first step involved identifying and selecting welfare assessment tools focusing on pre-weaned dairy calves. A total of 4 assessment tools focusing on the welfare of dairy calves during the preweaning period were identified (Vasseur et al., 2010; Barry et al., 2019; Hayer et al., 2021; and the Welfare Quality® assessment protocols for veal calves). These tools were developed to assess the welfare of dairy calves in Canada and Europe. Other assessment tools for specific outcomes focusing on dairy calves were identified. These tools were the calf hygiene score (Kellermann et al., 2020), a scoring system for the assessment of omphalitis (Steerforth and Van Winden, 2018), the fecal score proposed by McGuirk (2008), a respiratory score assessment tool (Poulsen and McGuirk, 2009), and the body mass index proposed by Scott et al. (2019). The authors suggested the assessment of dehydration by evaluating the presence of an ocular depression in the orbit and the cervical skin tent. The HRQoL tool (Bell et al., 2023) was not included in this selection due to its focus on BRD and because the tool was in the development phase.

All these tools were organized into a file, and a second step involving the selection of welfare indicators was carried out. This selection was performed by the 7 co-authors of this paper, veterinarians and animal scientists. Three criteria were used for the selection of the welfare indicators. The first was that welfare indicators could be assessed during punctual farm visits without expensive equipment (i.e., equipment costing less than a few hundred dollars). Considering the limited time consultants have during farm visits, the second criterion was that indicators could be assessed within 1 and 3 h for a maximum of 20 calves, accounting for the different areas related to the pre-weaned calves (i.e., from birth, including the calving area, the neonatal period, and until weaning). Some behavioral indicators, such as feeding behavior, were not included in this selection due to the time required for

evaluation. The third criterion was that, in situations with similar indicators, those with higher reliability were chosen according to the experts' perception. For example, the indicator 'Underdevelopment/Runt' characterizing calf body condition scoring was selected by Hayer et al. (2021), considering the following categories for evaluation: 'Score 0 = Good muscles, ribs not visible, shiny coat; Score 1 = Lack of muscles, ribs visible, dull coat.' The Welfare Quality® program selected a similar indicator called very lean animals, using the categories: 0 – Normal condition, 2 – Severe lower condition (approx. 30% less than average). These 2 ordinal scoring methods may pose reliability problems within and between raters. Another method was therefore considered by experts as less subjective, such as the utilization of the Body Mass Index proposed by Scott et al. (2019) by considering the weight, length from the withers to the lumbosacral junction, and hip height. This approach was also applied to other indicators when relevant. In addition, some indicators have been adjusted for a more objective assessment when necessary. Identifying reliable indicators would be easier if reliability were reported for the indicators in the selected tools.

A total of 44 on-farm calf welfare indicators were selected (Table 1). Of these, 21 were animal-based, and 23 indicators were resource and management-based.

Validity, reliability and feasibility

Various parameters need to be considered when selecting an on-farm indicator. This review will focus on welfare indicators' validity, reliability, and feasibility (Knierim and Winckler, 2009).

Validity was referred to as 'the measure measures what it is supposed to' (de Graaf et al., 2017). Recent assessment tools focusing on dairy calves highlighted that information on the validation of animal-based indicators is limited and needs further exploration (Barry et al., 2019; Hayer et al., 2021). In another assessment tool focusing on calves, the validation of resource- and management-based indicators was conducted by 6 experts from different fields related to dairy production (Vasseur et al., 2010). The expert group was responsible for examining and selecting the indicators. Another approach, using the opinions of trained users, was used to assess the validity of welfare indicators (including behavioral indicators) for dairy cows (de Graaf et al., 2017). The trained users judged the validity of the indicators using a scale ranging from 0 to 100 (not acceptable to very acceptable). The strategy of consulting expert opinion was used in our review to select welfare indicators for pre-weaned dairy calves (previous section). However, the selected indicators and welfare dimension could differ according to different approaches and groups of experts, as we dis-

Table 1. Description of calf welfare indicators at the preweaning period observed in the literature

Level	Welfare indicator*	Description	Reference
Animal-based	Demeanor	0 = bright, alert, responsive; 1 = Dull, possibly depressed, less responsive; 2 = Dull, apathetic, unresponsive.	Barry et al. (2019)
	Mobility	Ability to stand unassisted and move freely; 3-point scale of 0 to 2; 0 = Stands unassisted, actively mobile; 1 = Slow to stand, limited mobility; 2 = Assistance required to stand, no mobility.	Barry et al. (2019)
	Umbilical pain	The pain was assessed through palpation of the umbilical stump, and the elicited pain response — flinch and kyphosis.	Steerforth and Winden (2018)
	Avoidance distance/calf escape test (fear)	a = no eye contact; b = eye contact succeeded, step 1 results in a withdrawal; c = step 1 succeeded, step 2 results in a withdrawal; d = step 2 succeeded, calf cannot be touched; e = calf can be touched.	Welfare Quality® assessment protocols for veal calves
	Cleanliness of carpal joints	Score 0 = Clean carpal joints; Score 1 = Minor soiled carpal joints; Score 2 = Coarse soiling, wet carpal joints.	Hayer et al. (2021)
	Calf hygiene score (Belly)	Score 1 = 0 - 10%; score 2 = > 10–30%; score 3 = > 30%.	Kellermann et al. (2020)
	Calf hygiene score (Side)	Score 1 = 0 - 10%; score 2 = > 10–30%; score 3 = > 30%.	Kellermann et al. (2020)
	Calf hygiene score (Rear)	Score 1 = 0 - 10%; score 2 = > 10–30%; score 3 = > 30%.	Kellermann et al. (2020)
	Alopecia	Presence of alopecia and region.	Welfare Quality® assessment protocols for veal calves, adjusted.
	Lesions	Presence and type of lesion (claw lesions, joint lesions, bursae, hard skin spots, bitten tail/ear).	Welfare Quality® assessment protocols for veal calves
	Fecal score	0 = normal consistency; 1 = semi-formed or pasty; 2 = loss of consistency but enough consistency to stay on bedding; 3 = watery feces sieving through bedding material.	McGuirk (2008)
	Dehydration (Eyes and skin)	Ocular depression in the orbit and cervical skin tent.	Suggested by the authors
	Cough	0 = none; 1 = induce a single cough; 2 = Repeated cough induced or occasional spontaneous cough; 3 = repeated spontaneous coughs.	Poulsen and McGuirk (2009)
	Nasal discharge	0 = Normal serous discharge; 1 = Small amount of unilateral turbid discharge; 2 = bilateral, turbid, or excessive mucus discharge; 3 = Abundant bilateral muco-purulent discharge.	Poulsen and McGuirk (2009)
	Ocular discharge	0 = normal; 1 = small amount of ocular discharge; 2 = moderate amount of bilateral flow; 3 = heavy ocular discharge.	Poulsen and McGuirk (2009)
	Ear carriage	0 = normal, 1 = ear flick or head shake, 2 = slight unilateral droop, 3 = head tilt or bilateral droop.	Poulsen and McGuirk (2009)
	Rectal temperature	Objective assessment with the aid of a thermometer.	Poulsen and McGuirk (2009)
	Swollen navel - Omphalitis	Thickening of the umbilical region.	Steerforth and Winden (2018)
	Navel secretion	Pus or abscess on the umbilical stump.	Steerforth and Winden (2018)
	Umbilical hernia	A palpable umbilical hernia.	Steerforth and Winden (2018)
	Body mass index	Body weight is divided by the sum of the calf's length from the withers to the lumbosacral junction and the hip height.	Scott et al. (2019)

Continued

cussed previously. Another validation could be achieved by comparing the indicator with another, more objective measure. As an example, the use of fecal consistency scoring for evaluating diarrhea in calves, ranging from 0 (normal) to 3 (watery), was validated by considering

the fecal dry matter, a more objective method (Renaud et al., 2020). This approach could be used to validate other welfare indicators focusing on pre-weaned dairy calves.

The complexity of validating behavioral measures was emphasized in a previous study (Boateng et al., 2018),

Table 1 (Continued). Description of calf welfare indicators at the preweaning period observed in the literature

Level	Welfare indicator*	Description	Reference
Resource- and management-based	Colostrum feeding method	Bottle with nipple; Nipple pail, Open pail; Esophageal feeder; Other.	Vasseur et al. (2010)
	Colostrum equipment cleanliness	Very clean, acceptable or unacceptable.	Vasseur et al. (2010)
	Milk-feeding method	A. Bottle with nipple; B. Nipple pail; C. Open pail; D. Automatic delivery system. E. Large nipple bucket; F. Other.	Vasseur et al. (2010)
	Milk bucket/distributor cleanliness	Very clean, acceptable or unacceptable.	Vasseur et al. (2010)
	Access to water	Yes, or not.	Vasseur et al. (2010), adjusted.
	Type of waterers	Pail; Paddle or push-button water bowl; Automatic waterers	Vasseur et al. (2010), adjusted.
	Water cleanliness in calf waterers	Very clean, acceptable or unacceptable.	Vasseur et al. (2010)
	Access to forage	Yes, or not.	Vasseur et al. (2010), adjusted.
	Method of distribution of forage	A. Forage trough; B. On floor; C. Other.	Vasseur et al. (2010)
	Access to concentrate	Yes, or not.	Vasseur et al. (2010), adjusted.
	Method of distribution of concentrate	A. Concentrate container; B. On floor; C. Others.	Vasseur et al. (2010)
	Type of housing used for dairy calves	Individual/Group; Outside/Greenhouse/In the barn/Nursey; Pens with bars/Pens with solid walls/ Closed hutch/ Opened hutch/ and other.	Vasseur et al. (2010)
	Density (space available per calf)	Evaluation of housing dimensions (width and length) and the number of calves.	Vasseur et al. (2010)
	Bedding quantity in the calf area	A. Very plentiful, floor completely covered with bedding; B. Plentiful, most of the floor covered with bedding although bedding is thin at some places or mat and bedding; C. Scarce, most of the floor is visible; D. No bedding; E. No bedding.	Vasseur et al. (2010)
	Bedding cleanliness in the calf area	Very clean, acceptable or unacceptable.	Vasseur et al. (2010), adjusted.
	Bedding humidity in the calf area	Dry bed, partially wet bed, fully wet bed	Vasseur et al. (2010), adjusted.
	Walls cleanliness in the calf area	Very clean, acceptable or unacceptable.	Vasseur et al. (2010), adjusted.
	Calving pen	The presence or absence of a calving pen	Vasseur et al. (2010)
	Type of bedding in the calving pen	Identify the type of bedding for the calving pen.	Vasseur et al. (2010), adjusted.
	Bedding quantity in the calving pen	A. Very plentiful, floor completely covered with bedding; B. Plentiful, most of the floor covered with bedding although bedding is thin at some places or mat and bedding; C. Scarce, most of the floor is visible; D. No bedding; E. No bedding but mat.	Vasseur et al. (2010)
	Bedding humidity in the calving pen	Dry bed, partially wet bed, fully wet bed	Vasseur et al. (2010), adjusted.
	Bedding cleanliness in the calving pen	Very clean, acceptable or unacceptable.	Vasseur et al. (2010), adjusted.
	Walls cleanliness in the calving pen	Very clean, acceptable or unacceptable.	Vasseur et al. (2010), adjusted.

* We used different platforms, such as ScienceDirect and Scopus, to find assessment tools or protocols related to calf welfare assessments. Some indicators have been adjusted for a more objective assessment when necessary.

and best practices for developing and validating scales to assess behaviors, feelings, or actions were proposed. This process can be stratified into 9 steps, involving the identification of domains and item generation, content validity, pre-testing of questions, sampling and survey administration, item reduction, factor extraction, tests of dimensionality, tests of reliability, and tests of validity (Boateng et al., 2018). This approach could be used to develop and validate tools for assessing HRQoL for dairy calves.

Inter- and intra-rater reliability needs to be considered for welfare indicators. A short description of intra-rater reliability was reported in one of the selected tools (Barry et al., 2019). High intra-rater reliability was observed for

bedding quality assessments and clinical health assessments. The lack of information on inter-rater reliability is a limitation when considering different people's tool use. In another assessment tool, neither parameter was explored (Hayer et al., 2021). In the tool focusing on resource- and management-based indicators (Vasseur et al., 2010), inter-rater reliability was explored with 2 well-trained raters. High reliability was observed for qualitative environment-based indicators. Two indicators had moderate reliability (i.e., accumulation of manure and the quantity of litter in heifer housing).

The inter- and intra-rater reliability of the calf hygiene score was explored using pictures of approximately 40 Holstein calves (Kellermann et al., 2020). The calf hy-

giene score demonstrated high reliability, ranging from moderate to strong for inter- and intra-rater agreement. In another study, the inter-rater reliability of clinical signs used to diagnose BRD in veal calves was assessed among veterinarians, technicians, and producers (Berman et al., 2021). The intra-rater reliability for the selected indicators was variable, and differences were observed between the end-users. It was observed that reliability has not been explored extensively considering the selected assessment tools. This information highlights the importance of assessing the reliability (inter- and intra-rater) of health indicators and indicators from other areas of animal welfare.

Different points need to be considered when evaluating reliability: the selected indicator, the number of raters and their selection (especially to obtain the inter-rater reliability), the number of samples that will be evaluated, the number of repetitions (essential for obtaining intra-rater reliability), the used approach (e.g., on the farm, using pictures, or videos), the statistical method adopted, and how the results will be interpreted (Koo and Li, 2016). As a final consideration for the reliability test, although observer training was not included as a criterion for selecting the tools in the last section, it could be essential for achieving reliable results and on-farm application. For example, trained assessors' reliability was higher than that of non-trained assessors in evaluating body condition score for dairy cows (Vasseur et al., 2013). Future studies could explore the inter- and intra-rater reliability of welfare indicators for pre-weaned dairy calves.

Feasibility typically hinges on the time required to evaluate an indicator. For the assessment of dairy calf welfare, a total of 2.5 h per person was necessary to evaluate a minimum of 20 calves (requiring 2 people for assessment; Barry et al., 2019). Specifically, 1 h was allocated for behavior assessment, during which calves were filmed for subsequent evaluation. Despite selecting the assessment tool proposed by Barry et al. (2019), we did not consider some behavioral indicators feasible for data collection during a farm visit. This is one of the reasons why we did not select many behavioral indicators. Furthermore, the selected behavioral expressions were limited to the tools focused on assessing calf welfare, which already presents a limitation in selecting behavioral indicators. This review could be extended in future studies, and more behavioral indicators could be selected and validated for assessing calf welfare and HRQoL. More information on the evaluation time of behavioral indicators may be necessary to include these indicators in feasible welfare and HRQoL assessment tools.

For the Canadian approach, which involves management- and resource-based indicators, a maximum of 3 h was allocated for data collection (Vasseur et al., 2010). While the authors deemed the estimated time sufficient

to consider the developed tools feasible, another critical aspect is whether the potential users will find the selected indicators useful for the assessment tool. This assessment was made in studies focusing on the welfare of dairy cows (Lievaart and Noordhuizen, 2011) and the HRQoL tool for calves (Bell et al., 2023). The usefulness of welfare indicators for pre-weaned dairy calves according to potential users could be explored in more detail.

Another important aspect of feasibility is the cost of equipment to assess specific indicators. Low-cost indicators are important for on-farm assessment tools. However, not only the cost of the measure must be considered; the economic importance of the assessment tools, considering different indicators, also needs to be explored. Reporting a possible association between the calf welfare at the farm level (using one of the assessment tools) and the performance of dairy calves and the price of surplus calves sold could be important for motivating producers and potential users to adopt the assessment tools.

CONCLUSIONS

This review integrates animal welfare and HRQoL concepts for pre-weaned dairy calves. HRQoL, as a component of the QoL concept, is used to determine how an animal feels during illness using behavioral expressions of affective states. We adapted a human HRQoL framework for dairy calves, illustrating the interconnection of 3 domains of animal welfare (behavior, mental state and health) in assessing calves' HRQoL. It is worth noting that a limited number of HRQoL assessment tools have been developed for dairy calves, and there is no standard way to assess the welfare of pre-weaned dairy calves. Some studies focus on specific aspects of animal welfare, while others address this concept more broadly. While behavioral indicators have been explored in dairy science, they often remain disconnected from the concept of HRQoL. Considering the selected welfare indicators for pre-weaned dairy calves, we observed that validity, reliability, and feasibility have not been extensively explored in the literature. This review contributes to the understanding of welfare and HRQoL concepts and their assessment.

ACKNOWLEDGMENTS

The authors acknowledge the student funding support provided by the Regroupement Op+lait (Saint-Hyacinthe, Canada), the Faculté de Médecine Vétérinaire, Université de Montréal (Saint-Hyacinthe, Canada), the Études Supérieures et Postdoctorales (ESP) at Université de Montréal (Montréal, Canada), the Innov'Action Programme (Ministère de l'Agriculture, des Pêcheries et de l'Alimentation, Québec, Canada), the Producteurs

de Bovins du Québec (Québec, Canada), and the Groupe de Recherche en Santé Bovine (GRESABO; Saint-Hyacinthe, Canada). The authors have not stated any conflicts of interest.

REFERENCES

- Barry, J., E. Kennedy, R. Sayers, I. J. M. De Boer, and E. A. M. Bokkers. 2019. Development of a welfare assessment protocol for dairy calves from birth through to weaning. *Anim. Welf.* 28:331–344. <https://doi.org/10.1120/09627286.28.3.331>.
- Bell, D. J., C.-A. Duthie, C. S. Mason, A. Hancock, C. Penny, I. Odeyemi, D. J. Bartram, and M. J. Haskell. 2024. Developing a tool to assess the health-related quality of life in calves with respiratory disease: tool refinement and construct validity testing. *Animal* 17:101215. <https://doi.org/10.1016/j.animal.2024.101215>.
- Bell, D. J., B. Vigers, C. A. Duthie, D. J. Bartram, A. Hancock, I. Odeyemi, C. Penny, and M. J. Haskell. 2023. Developing a tool to assess the health-related quality of life in calves with respiratory disease: content validation. *Animal* 17:100702. <https://doi.org/10.1016/j.animal.2022.100702>.
- Berman, J., D. Francoz, A. Abdallah, S. Dufour, and S. Buczinski. 2021. Evaluation of inter-rater agreement of the clinical signs used to diagnose bovine respiratory disease in individually housed veal calves. *J. Dairy Sci.* 104:12053–12065. <https://doi.org/10.3168/jds.2021-20503>.
- Blokhuys, H. J., I. Veissier, M. Miele, and B. Jones. 2010. The welfare quality® project and beyond: Safeguarding farm animal well-being. *Acta Agric. Scand. A Anim. Sci.* 60:129–140. <https://doi.org/10.1080/09064702.2010.523480>.
- Boateng, G. O., T. B. Neillands, E. A. Frongillo, H. R. Melgar-Quinonez, and S. L. Young. 2018. Best Practices for Developing and Validating Scales for Health, Social, and Behavioral Research: A Primer. *Front. Public Health* 6:149. <https://doi.org/10.3389/fpubh.2018.00149>.
- Bull, E. M., D. J. Bartram, B. Cock, I. Odeyemi, and D. C. J. Main. 2021. Construction of a conceptual framework for assessment of health-related quality of life in calves with respiratory disease. *Animal* 15:100191. <https://doi.org/10.1016/j.animal.2021.100191>.
- Conboy, M. H., C. B. Winder, C. Medrano-Galarza, S. J. LeBlanc, D. B. Haley, J. H. C. Costa, M. A. Steele, and D. L. Renaud. 2021. Associations between feeding behaviors collected from an automated milk feeder and disease in group-housed dairy calves in Ontario: A cross-sectional study. *J. Dairy Sci.* 104:10183–10193. <https://doi.org/10.3168/jds.2021-20137>.
- Cramer, M. C., T. L. Ollivett, and A. L. Stanton. 2016. Associations of behavior-based measurements and clinical disease in preweaned, group-housed dairy calves. *J. Dairy Sci.* 99:7434–7443. <https://doi.org/10.3168/jds.2015-10207>.
- Cramer, M. C., K. L. Proudfoot, and T. L. Ollivett. 2019. Short communication: Behavioral attitude scores associated with bovine respiratory disease identified using calf lung ultrasound and clinical respiratory scoring. *J. Dairy Sci.* 102:6540–6544. <https://doi.org/10.3168/jds.2018-15550>.
- de Graaf, S., B. Ampe, C. Winckler, M. Radeski, L. Mounier, M. K. Kirchner, M. J. Haskell, F. J. C. M. van Eerdenburg, A. B. des Roches, S. N. Andreasen, J. Bijttebier, L. Lauwers, W. Verbeke, and F. A. M. Tuytens. 2017. Trained-user opinion about Welfare Quality measures and integrated scoring of dairy cattle welfare. *J. Dairy Sci.* 100:6376–6388. <https://doi.org/10.3168/jds.2016-12255>.
- Farm Animal Welfare Council. 1993. Second Report on Priorities for Research and Development in Farm Animal Welfare. Londres: DEFRA.
- Fraser, D. 2008. Understanding animal welfare. *Acta Vet. Scand.* 50(S1):S1. <https://doi.org/10.1186/1751-0147-50-S1-S1>.
- Fraser, D., D. M. Weary, E. A. Pajor, and B. N. Milligan. 1997. A scientific conception of animal welfare that reflects ethical concerns. *Anim. Welf.* 6:187–205. <https://doi.org/10.1017/S0962728600019795>.
- Griebel, P., K. Hill, and J. Stookey. 2014. How stress alters immune responses during respiratory infection. *Anim. Health Res. Rev.* 15:161–165. <https://doi.org/10.1017/S1466252314000280>.
- Guevara-Mann, D., D. L. Renaud, and M. C. Cantor. 2023. Activity behaviors and relative changes in activity patterns recorded by precision technology were associated with diarrhea status in individually housed calves. *J. Dairy Sci.* 106:9366–9376. <https://doi.org/10.3168/jds.2023-23380>.
- Guyatt, G. H., C. E. Ferrans, M. Y. Halyard, D. A. Revicki, T. L. Symonds, C. G. Varricchio, A. Kotzeva, J. M. Valderas, and J. L. Alonso. 2007. Exploration of the value of health-related quality-of-life information from clinical research and into clinical practice. *Mayo Clin. Proc.* 82:1229–1239. <https://doi.org/10.4065/82.10.1229>.
- Hampton, J. O., L. M. Hemsworth, P. H. Hemsworth, T. H. Hyndman, and P. Sandøe. 2023. Rethinking the utility of the Five Domains model. *Anim. Welf.* 32:e62. <https://doi.org/10.1017/awf.2023.84>.
- Harvey, N. D. 2021. A Simple Guide to Inter-rater, Intra-rater and Test-retest Reliability for Animal Behaviour Studies. <https://doi.org/10.31219/osf.io/8stpy>.
- Hayer, J. J., D. Nysar, C. Heinemann, C. D. Leubner, and J. Steinhoff-Wagner. 2021. Influences on the assessment of resource- and animal-based welfare indicators in unweaned dairy calves for usage by farmers. *J. Anim. Sci.* 99:skab266. <https://doi.org/10.1093/jas/skab266>.
- Hulbert, L. E., and S. J. Moisa. 2016. Stress, immunity, and the management of calves. *J. Dairy Sci.* 99:3199–3216. <https://doi.org/10.3168/jds.2015-10198>.
- Kellermann, L. M., A. Rieger, G. Knubben-Schweizer, and M. Metzner. 2020. Short communication: Design and validation of a hygiene score for calves. *J. Dairy Sci.* 103:3622–3627. <https://doi.org/10.3168/jds.2019-17536>.
- Knierim, U., and C. Winckler. 2009. On-farm welfare assessment in cattle: Validity, reliability and feasibility issues and future perspectives with special regard to the Welfare Quality® approach. *Anim. Welf.* 18:451–458. <https://doi.org/10.1017/S0962728600000865>.
- Koo, T. K., and M. Y. Li. 2016. A Guideline of Selecting and Reporting Intraclass Correlation Coefficients for Reliability Research. *J. Chirop. Med.* 15:155–163. <https://doi.org/10.1016/j.jcm.2016.02.012>.
- Lago, A., S. M. McGuirk, T. B. Bennett, N. B. Cook, and K. V. Nordlund. 2006. Calf respiratory disease and pen microenvironments in naturally ventilated calf barns in winter. *J. Dairy Sci.* 89:4014–4025. [https://doi.org/10.3168/jds.S0022-0302\(06\)72445-6](https://doi.org/10.3168/jds.S0022-0302(06)72445-6).
- Lievaart, J. J., and J. P. T. M. Noordhuizen. 2011. Ranking experts' preferences regarding measures and methods of assessment of welfare in dairy herds using Adaptive Conjoint Analysis. *J. Dairy Sci.* 94:3420–3427. <https://doi.org/10.3168/jds.2010-3954>.
- Louie, A. P., J. D. Rowe, W. J. Love, T. W. Lehenbauer, and S. S. Aly. 2018. Effect of the environment on the risk of respiratory disease in preweaning dairy calves during summer months. *J. Dairy Sci.* 101:10230–10247. <https://doi.org/10.3168/jds.2017-13716>.
- Lund, V., G. Coleman, S. Gunnarsson, M. C. Appleby, and K. Karkinen. 2006. Animal welfare science - Working at the interface between the natural and social sciences. *Appl. Anim. Behav. Sci.* 97:37–49. <https://doi.org/10.1016/j.applanim.2005.11.017>.
- McGuirk, S. M. 2008. Disease Management of Dairy Calves and Heifers. *Vet. Clin. North Am. Food Anim. Pract.* 24:139–153. <https://doi.org/10.1016/j.cvfa.2007.10.003>.
- Mellor, D. J., N. J. Beausoleil, K. E. Littlewood, A. N. McLean, P. D. McGreevy, B. Jones, and C. Wilkins. 2020. The 2020 five domains model: Including human-animal interactions in assessments of animal welfare. *Animals (Basel)* 10:1–24. <https://doi.org/10.3390/ani10101870>.
- Mellor, D. J., and C. S. W. Reid. 1994. Concepts of Animal Well-Being and Predicting the Impact of Procedures on Experimental Animals. Improving the well-being of animals in the research environment, 3–18.
- Poulsen, K. P., and S. M. McGuirk. 2009. Respiratory Disease of the Bovine Neonate. *Vet. Clin. North Am. Food Anim. Pract.* 25:121–137. <https://doi.org/10.1016/j.cvfa.2008.10.007>.
- Reimert, I., L. E. Webb, M. A. van Marwijk, and J. E. Bolhuis. 2023. Review: Towards an integrated concept of animal welfare. *Animal* 100838:100838. <https://doi.org/10.1016/j.animal.2023.100838>.
- Renaud, D. L., L. Buss, J. N. Wilms, and M. A. Steele. 2020. Technical note: Is fecal consistency scoring an accurate measure of fecal dry

- matter in dairy calves? *J. Dairy Sci.* 103:10709–10714. <https://doi.org/10.3168/jds.2020-18907>.
- Scott, K., D. F. Kelton, T. F. Duffield, and D. L. Renaud. 2019. Risk factors identified on arrival associated with morbidity and mortality at a grain-fed veal facility: A prospective, single-cohort study. *J. Dairy Sci.* 102:9224–9235. <https://doi.org/10.3168/jds.2019-16829>.
- Steerforth, D. D., and S. Van Winden. 2018. Development of clinical sign-based scoring system for assessment of omphalitis in neonatal calves. *Vet. Rec.* 182:549. <https://doi.org/10.1136/vr.104213>.
- Testa, M. A., and D. C. Simonson. 1996. Assessment of quality-of-life outcomes. *N. Engl. J. Med.* 334:835–840. <https://doi.org/10.1056/NEJM199603283341306>.
- Vasseur, E., J. Gibbons, J. Rushen, and A. M. De Passille. 2013. Development and implementation of a training program to ensure high repeatability of body condition scoring of dairy cows. *J. Dairy Sci.* 96:4725–4737. <https://doi.org/10.3168/jds.2012-6359>.
- Vasseur, E., J. Rushen, A. M. de Passillé, D. Lefebvre, and D. Pellerin. 2010. An advisory tool to improve management practices affecting calf and heifer welfare on dairy farms. *J. Dairy Sci.* 93:4414–4426. <https://doi.org/10.3168/jds.2009-2586>.
- Webb, L. E., R. Veenhoven, J. L. Harfeld, and M. B. Jensen. 2019. What is animal happiness? *Ann. N. Y. Acad. Sci.* 1438:62–76. <https://doi.org/10.1111/nyas.13983>.
- Yeates, J. 2016. Quality of life and animal behaviour. *Appl. Anim. Behav. Sci.* 181:19–26. <https://doi.org/10.1016/j.applanim.2016.04.018>.

ORCIDS

- Jean Ramos,  <https://orcid.org/0000-0002-9159-3563>
- Marianne Villettaz-Robichaud,  <https://orcid.org/0000-0001-9685-1827>
- Jocelyn Dubuc,  <https://orcid.org/0000-0003-1013-8372>
- Débora Santschi,  <https://orcid.org/0000-0002-3938-6073>
- Jean-Philippe Roy,  <https://orcid.org/0000-0002-0444-2303>
- Sébastien Buczinski  <https://orcid.org/0000-0002-8460-4885>